AMA 572: Credit Risk
Midterm Examination

Examination Questions

Time: 150 minutes **Total:** 100 points

Show all essential steps. Unless otherwise stated, interest rates are continuously compounded, all expectations used for pricing are under the risk-neutral measure, and Φ denotes the standard normal cumulative distribution function. A non-programmable calculator and a standard normal table may be used.

Examination Questions

Question 1. Rating transitions and generator estimation [15 points]

There are three credit states: A, B, and default D. Default is absorbing. At time 0 there are 12 firms in state A and 8 firms in state B. During the following year:

- At month 3, two A-rated firms migrate to B.
- At month 6, one B-rated firm migrates to A, and one other B-rated firm defaults.
- At month 9, one A-rated firm defaults.
- No other transitions occur.

(a) Compute the empirical one-year transition matrix.

(b) Compute the total exposure times N_A and N_B, measured in firm-years.

(c) Estimate the generator $\widehat{\Lambda}$ using $\widehat{\lambda}_{ij} = M_{ij}/N_i$ for $i \neq j$.

(d) Compute the generator-based one-year transition matrix $\widehat{P}_{\text{gen}} = \exp(\widehat{\Lambda})$.

Question 2. Implied default probability from a bond price [10 points]

A one-year corporate bond has par value 100 and pays 104 at maturity if no default occurs. Its continuously compounded yield is 5%. The corresponding risk-free rate is 3%. Default can occur only at time 0.5, and recovery is 40 at that time.

Using the expected-loss method from the lectures, compute the implied probability p of default at time 0.5. In particular, compare the corporate and default-free bond prices and discount the loss given default from time 0.5 to time 0.

Question 3. Merton's structural model [20 points]

In Merton's model, a firm has current asset value $V_0 = 100$. It has one zero-coupon debt issue with face value $D = 80$ and maturity $T = 1$ year. Assume $r = 5\%$, asset volatility $\sigma_V = 30\%$, no payouts, no taxes, and no bankruptcy costs.

(a) Explain why equity is a European call option on the firm's assets, and write its value.

(b) Compute d_1, d_2, the equity value S_0, and the debt value B_0.

(c) Compute the risk-neutral probability of default by maturity.

(d) Compute the continuously compounded risky debt yield and the credit spread.

Question 4. First passage time and the Black-Cox model [15 points]

Under the risk-neutral measure, let

$$\frac{\mathrm{d}V_t}{V_t} = (r-\delta)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \tau = \inf\{t \geq 0 : V_t \leq V_B\},$$

where $0 < V_B < V_0$. Define

$$\mu = r - \delta - \frac{1}{2}\sigma^2, \qquad b = \log\left(\frac{V_B}{V_0}\right) < 0.$$

(a) State the Laplace transform $\mathbb{E}[e^{-s\tau}]$ for $s > 0$.

(b) Show that the probability of default by time T is

$$\mathbb{P}(\tau \leq T) = \Phi\left(\frac{b-\mu T}{\sigma\sqrt{T}}\right) + \exp\left(\frac{2\mu b}{\sigma^2}\right)\Phi\left(\frac{b+\mu T}{\sigma\sqrt{T}}\right).$$

(c) Evaluate this probability when $V_0 = 100$, $V_B = 70$, $r = 4\%$, $\delta = 2\%$, $\sigma = 25\%$, and $T = 2$.

Question 5. Leland-Toft endogenous default [15 points]

In the Leland-Toft model, new debt with par value $p\,\mathrm{d}t$ is issued continuously. Each bond has an exponentially distributed maturity with density $\phi(u) = me^{-mu}$, so the mean maturity is $1/m$. The total coupon rate per unit par is ρ, the corporate tax rate is κ, and bondholders recover a fraction α of firm value at default.

(a) Show that the stationary total par value of outstanding debt is $P = p/m$.

(b) Suppose an individual bond with unit face value and maturity $T > \tau$ receives $ce^{-r(T-\tau)}$ at default. Show that matching aggregate recovery to αV_B gives

$$c = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

(c) Let

$$\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}, \qquad \gamma_r = \sqrt{\xi^2 + \frac{2r}{\sigma^2}} + \xi,$$

so that $\mathbb{E}[e^{-r\tau}] = (V_B/V)^{\gamma_r}$. Derive the total firm value

$$v(V; V_B) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^{\gamma_r}.$$

(d) Briefly explain the two-stage optimization that determines the endogenous default barrier and the optimal debt level.

Question 6. Duffie-Singleton reduced-form pricing [10 points]

Suppose a defaultable claim recovers a fraction R_M of its market value immediately before default. The risk-free rate r and default intensity λ are constant.

(a) Give a short discrete-time argument showing that the effective discount rate is $r + (1 - R_M)\lambda$.

(b) Hence price a four-year defaultable zero-coupon bond with unit face value when $r = 3\%$, $\lambda = 2\%$, and $R_M = 40\%$. Compute its yield and credit spread.

Question 7. Five-year CDS valuation [15 points]

Consider a five-year CDS with unit notional. Premiums are paid annually at times $1, 2, 3, 4, 5$. The conditional probability of default in each year, given survival to the start of that year, is $q = 2.5\%$. Default, if it occurs in year k, occurs at time $k - 0.5$. The continuously compounded risk-free rate is $r = 4\%$, and the recovery rate is $R = 35\%$.

The premium buyer pays the regular premium s at year-end if the reference entity survives, and pays one-half year's accrued premium if default occurs halfway through a year. Compute the break-even annual CDS spread s in basis points by equating the premium and protection legs.